



Gaurav Rajendra Mahajan
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|Harness (Mechanical) Design Engineer | Routing & Packaging Expert
Location Preference: Pune, Mumbai, USA (H1B)

Profile Summary

- A result-oriented professional offering **over 11 years** of successful career distinguished by commendable performance in **design and development of Automobile harness (6.4 years), Heavy Engines(2.6 years) and Torpedo Launching System (2 years)**.
- Extensive knowledge of complete harness concept design to release it, get parts on contract & push the new design on assembly line for full phase production.
- Extensive knowledge of tools like **CAD (Nx) software, Teamcenter for PLM, aPriority for costing, Auros for knowledge management, Reliasoft for FMEA & DRBFM**.
- Authorized to work in **Illinois, US state under H1B visa**
- Successful leading various projects on chassis vehicle subsystem for various geographies like North America, Canada, Mexico.
- Skilled in managing complete engineering operations involving requirement analysis, specifications, designing, prototype development, testing and development in compliance to global quality standards
- Resolve difficult assembly problems by simplifying, common approach & writing standard processes, conduct DFMEA and cross functional design reviews for new design
- Proficient in leading full **product lifecycle activities** including concept maturation, design management, engineering change management, issues control & resolution, cost management, various other cross-functional activities in designing and developing products
- Effective leader with excellent motivational skills to sustain growth momentum while motivating team performance

Core Competencies

Project Management	Change Management	Leadership	Product Development
Value Engineering	Product Design Engineering	Process Excellence	FMEA & DRBFM
Reverse Engineering	CAD development	Team Management	Risk Mitigation
Harness Design	Component packaging	Supplier communication	Cross Functional Review

Organizational Experience

Company Name: Tech Mahindra Limited. (3rd December 2018-Present)

Company Name: Eteam InfoServices Pvt Ltd. (26th December 2017-2nd December 2018)

Project Name: US Truck chassis Electrical Harnessing [from June18-Present]

Working as program lead for Chassis wiring & battery cables for trucks & buses. My area of work includes Leading a project from the starting phase to complete production deployments at the production line. This Includes **gathering & finalizing the program inputs** provide the **change in cost projection, Design hours requirement, Lead availability time for parts**. Define the requirement of DRBFM & DFMEA, Design validation plan, communicating requirements with suppliers, Purchase team. Design chassis wire harness & battery cables. Propose its packaging on all possible configurations. Verify the design with schematic / system engineers. Release the design with quality in system. Resolve the production issues with ICA & PCA.

Design support due to other system changes like new engines 2021, 2025, 2027, GW transmission etc. Currently working on development of new features like **ADAS system, TFR, new aero sleeper cab, common engine powertrain** etc. Also working on development of new **24V architecture**.

Job Responsibility:-

- Understand project requirements & plan the target dates with **key deliverables**.
- Find & propose the **cost impact** for a given change including cost of tooling, part procurement, and design hours requirement.
- Define & plan **DRBFM & DFMEA** as required.

- Use of Auros IQ to use existing knowledge & **create an intellectual property**. This includes releasing new designs with standards & quality, **developing new standards (K-packs)** after DFMEA, DRBFM of new function/design.
- Define & execute Design validation plan (**DVP**).
- Review the release work as a quality check.
- **3D concept design** for battery cables & chassis harness for packaging prospective.
- Propose correct wire type, lengths & parts, complete harness BOM.
- Finding a best cost by **estimating with aPriori tool** by comparing various manufacturing methods & locations.
- Selection of gauge of wire by **calculating voltage drop**.
- Creation of **installation drawings, part drawing with GD&T**.
- Track job order & make sure design release & parts will be available before its build date. Make it sure that new design will be ready on various build evens (Concept vehicle, Display vehicle, certification, full production)
- Solve assembly problems by responding to **DCR**, propose **ICA & PCA**.
- Achieved non-negotiable milestones with parallel developments, contingency plans, resource balancing.
- Plan, schedule to integrate into main project timeline, cross functional team requirement integrated into plan. Ensure team Collaboration to deliver the project successfully on time

Company Name: Stratosphere IT Services Pvt Ltd. (17th April 2015-22nd December 2017)

Project Name: Engine Automation & Electrical Harnessing (June15-December17)

Working **onsite at L&T TS** on Automaton systems for marin(heavy) Engines. W26 are compact & light units with low operating costs, also used for generator set application.

Job Responsibility:-

- Develop the new systems for 12V26 & 16V26 with standard & optional kits with **cost down 15% approach**.
- **Selection of sensors, cable glands** & its related part as per assembly requirements from the approved suppliers.
- **Routing electrical path** considering minimum & easy maintenance point of view.
- Creation of **harness drawing (Form board Drawing)** indicating actual & cut lengths of cables, bill of materials etc.
- Creation of **assembly drawing** for harness indicating full routing path.
- **Design of supporting structure** for sensors & cables.

Proposal for how to use NX electrical harnessing module:-[2 months]

- Study the NX routing module and apply the software technology to existing working processes.
- Qualifying standard parts by adding built in paths, fixing & multi-ports.
- Proposal for formboard drawing & packaging of complete harness as used in automobile industry. (NX 9 with legacy file)
- Study of variant rule & variant configuration to create arrangements / configurations in harness.

Value Analysis / Value Engineering [Va/Ve] for W50DF & W46F:-

To suggest modification in approach of Va/Ve I studied the following topics.

- Concept of "Intelligent Engines". These engines do not need to be opened for checking of parts. The automation system is designed so that it indicates the main part's health.
- Possible solutions to reduce NOx emission. Such that using of two stage turbocharger, direct insert of water vapor in cylinder
- Studied new technology of Bearing Temperature Monitoring System.

Design of Insulation box for Turbocharger: - [4 months]

Wartsila updated their turbocharger with KBB ST5 & KBB ST6, so we need to modify the existing design of insulations for the newer one. Need to design for new cost down approach.

Job Responsibility:-

- Complete design of new insulations for turbochargers with the help of **Sheet metal** components.

- Creation of **technical drawings, CAD data, Models** & maintain in TC 10.
- Selection of manufacturing process like **Joining Methods, Painting instructions, Material specification, Insulation specification.**

Company name: DesignTech Systems ltd. (18th March 2013-15th April 2015).

Project name: TWC (Torpedo Weapon Charger)

This project has 2 phases each phase have 6 tubes. Each tube is unique due to position & orientation. Each single tube contains 12 mechanisms. Responsible to develop linkage/ operating mechanism manage all these CAD Data including models and drawings. Each mechanism & sub-mechanism contains arrangements.

Job Responsibility:-

Primary selection of **Fit, Tolerance & Welding**, Maintain DIN for Assembly, Collect the standard brought out parts step **model form vendors (sensors, plug, indicator)**, Check DCR & NCR and clear them, Update CAD model as per latest design, Create **manufacturing drawings** revise them as per new updates, Provide **assembly wise C.G., weight, Section Inertia details** as per required to design team, **Selection of fasteners** as per requirements.

Software skill:-

- Modelling & drafting (mostly fabrication).
- Arrangements for Main assembly & sub- assembly.
- Highly use Reference Sets option to optimize assembly performance.
- Advance use of Layers to demonstrate required parts of the model/assembly.
- Simple electrical & mechanical Routing.

Assembly Fixture for TWC:-

Design the assembly fixture for TWC to assemble various components & mechanisms on job. The current design is developed for shop & transportation purposes.

Job Responsibility:-

Selection of structure components according to load & stress developed at particular location. Check the fixture for testing purposes. The job will be accessible & the mechanism will work without any interface. Creation of drawings, BOM, etc. to hand over to shop.

UGNx Customization projects:

Automatic Attribute from Excel (UGNx) (Duration:-25 days):

This is a small customize command which when executed asks for an excel file & adds all attributes to the assembly component. The part is open to make display part environment & add attributes. This current command is done by a journal in VB language.

L&T Logo with single button (UGNx) (Duration:-30 days):

This is also a custom command which when executed creates a L&T logo on the required face in required scale also we need to give reference points for logo position. This is achieved with UDF (User Defined Feature).

Apply Annotation setting to existing old drawings: - (5 days):-

Currently working on this to apply the standard setting of colour, symbol & all annotation settings.

Knowledge Purview

3D CAD, NX UNIGRAPHICS: -

Electrical Automaton & Harnessing, Expression based Modelling, Drafting with GD & T, Assembly, sheet metal, Hydraulic & Mechanical routing, Fabrication, part family, UDF (User Defined Feature), Creation of UDF, and Clearance Analysis.

PLM, TeamCenter:

Use of team center, workflow, structure manager, Advance search, Impact analysis, change manager, use of substitute & alternator, use of Variant.

RPT (Rapid Prototype Training):

Orientation of given model, creation of .CMD file by using **CATALYST & INSIGHT 8.0**, Editing of layers in Insight for best result.

Working with Standards:-

Working with standards according to companies like ISO, ANSI Metric, ANSI Inches, Selection of Standard Metal Parts according to jobs like C - section, I - section, Nut & Bolts, Threads etc.

Programming Language:-VB, PHP, Java, JavaScript

Other softwares – **high level of Excel, aPriory, Auros, Reliasoft**

Academic Details

- Bachelor of Engineering in Mechanical from J.T.M College of Engineering, NMU University, Jalgaon, Maharashtra in 2012 with 63 % (First class).
- Higher Secondary School Certificate in Science from N.G. Patil College, Savda, Maharashtra in 2007 with 70%.
- Project - "Design & fabrication of Transporting & Lifting Machine"
- Seminar- "Ground Effect Aerodynamics of Race Cars"

Personal Details

Name	:	Gaurav R Mahajan.
Date of Birth	:	15th Feb 1990
Nationality	:	Indian.
Gender	:	Male.
Marital status	:	Married.
PASSPORT NO	:	U9139122
Languages	:	English, Hindi, Marathi
Current location	:	Chicago, Illinois, USA
Visa Status	:	H1B (valid up to 30/10/2024)
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